14.24 Solution: Comparing the form of Liénard-Wiechert potentials to the form of static results, we can see that there is a difference in the factor of

$$\frac{dt}{dt'} = 1 - \mathbf{n}\cdot\boldsymbol{\beta}.$$

This suggests that we should consider an integral along the loop in order to cancel this factor.

The circuit can be divided into N intervals of length Δ, so that there is only one particle in each interval at any instant. At time t, the scalar potential can be written as the total contribution from these intervals,

$$\Phi(\mathbf{x},t) = \sum_{k=1}^{N} \frac{q}{(1-\mathbf{n}_k(t_{\text{ret},k})\cdot\boldsymbol{\beta}_k(t_{\text{ret},k}))R_k},$$

where the retarded time is $t_{\text{ret},k} = t - R_k/c$, with $\mathbf{x}$ and $\mathbf{x}'_k$ as the location of the observer and the k-th interval, respectively, and $R_k = |\mathbf{x} - \mathbf{x}'_k(t_{\text{ret},k})|$.

We now want to turn the above summation to an integral. To this end, we have the equivalent expression

$$\Phi(\mathbf{x},t) = \frac{q}{\Delta}\sum_{k=1}^{N} \frac{\Delta}{(1-\mathbf{n}_k(t_{\text{ret},k})\cdot\boldsymbol{\beta}_k(t_{\text{ret},k}))R_k}.$$

In the limit of $N\to\infty$, every interval can be viewed as a straight line segment,

$$\Delta \approx v\delta t,$$

where δt is the time in the lab frame for a particle to traverse the k-th interval, and is related to the retarded time interval on the k-th interval by

$$\frac{\delta t}{\delta t_{\text{ret},k}} \approx 1 - \mathbf{n}_k(t_{\text{ret},k})\cdot\boldsymbol{\beta}_k(t_{\text{ret},k}).$$

Therefore,

$$\Phi(\mathbf{x},t) = \frac{q}{\Delta}\sum_{k=1}^{N}\frac{v\delta t_{\text{ret},k}}{R_k} \to \frac{q}{\Delta}\oint\frac{v dt_{\text{ret}}}{|\mathbf{x}-\mathbf{x}'(t_{\text{ret}})|},$$

in the $N\to\infty$ limit. Finally, we can relabel the time stamp for each inteval with the non-retarded time, as there is a one-to-one correspondence. Define the linear charge density as $\rho = q/\Delta$ and the length element $dl = vdt$, we will have

$$\Phi(\mathbf{x},t) = \frac{q}{\Delta}\oint\frac{vdt}{|\mathbf{x}-\mathbf{x}'(t)|} = \oint\frac{\rho dl}{|\mathbf{x}-\mathbf{x}'|}.$$

Therefore, for a steady current, the scalar potential is identical to the static result. We can obtain the same result for vector potential following the same argument. Then, for a steady current, there will be no radiation.